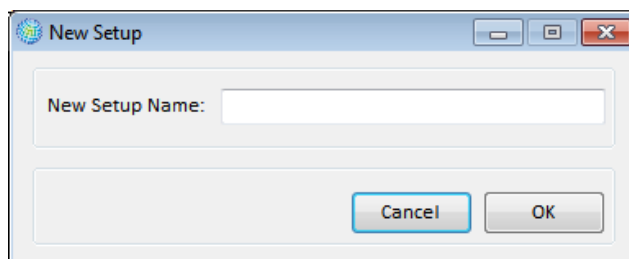


Add a Setup. To add a setup (e.g., to add a new country that was not pre-loaded with BenMAP-CE), click the **Add** button. The **New Setup** window will appear where you can type a name for the new setup.



After naming the new setup, you can define the elements that comprise a setup. Table 4-1 lists the 10 dataset types within BenMAP-CE and indicates which types of data are needed to perform certain analyses.

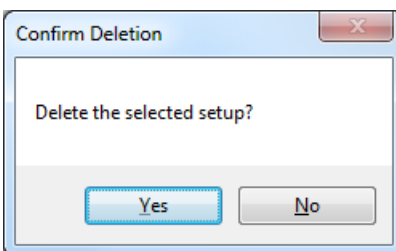
Table 4-1. BenMAP-CE Data Elements

Dataset Type	Required to Estimate Health Impacts	Required to Quantify Economic Values
Grid Definitions	✓	✓
Pollutants	✓	✓
Monitor Datasets (or Modeled Data)	✓	✓
Incidence/Prevalence Rates	✓	✓
Population Datasets	✓	✓
Health Impact Functions	✓	✓
Variable Datasets		
Inflation Datasets		✓
Valuation Functions		✓
Income Growth Adjustments		✓

Some of the elements of a setup are fundamental and should be entered before the others, namely, **Grid Definitions** and **Pollutants**. The **Incidence/Prevalence Rates**, **Population**, and **Variable Datasets** depend on the **Grid Definitions**, and the **Monitor** and **Health Impact Functions** datasets depend on the **Pollutants** that you have defined. Therefore, it is best to start by defining your **Grid Definitions** and **Pollutants**, and then define the other elements of the setup.

Modify a Setup. To modify a setup, click **Modify Datasets** on the menu bar. Choose the setup for modification from the drop-down list of **Available Setups** (the default value is *United States*). Then, click on the **Manage** button under one of the ten components comprising a setup. The sections below provide more information for each of these components.

Delete a Setup. To delete a setup, click **Modify Datasets** on the menu bar. Choose the setup for deletion from the drop-down list of **Available Setups**. Click the **Delete** button. You will then be asked to confirm your decision.



4.1.1 Grid Definitions

Fundamental Concept – Grid Definition

BenMAP-CE **Grid Definitions** help to spatially allocate data used in the benefits assessment. They enable users to conduct analyses with data sets at a variety of spatial scales and to aggregate and map results at different scales.

There are two types of grid definitions: **Shapefile Grid Definitions** and **Regular Grid Definitions**. Typically, a Shapefile Grid Definition is used when the areas of interest are political boundaries with irregularly shaped borders, while a Regular Grid Definition is used when the areas of interest are uniformly shaped grids (e.g., rectangles). All grid definitions must have a unique (i.e., non-repeating) column and row index.

A BenMAP-CE **Grid Definition** specifies geographic units (i.e., grid cells) that serve two purposes: (1) the program assigns air quality data, population data, baseline incidence rates, and health impact functions to these grids in order to calculate impacts; and (2) you can use these grids to report results in the GIS tool. You can define a grid in one of two ways: by loading a Shapefile (a particular type of GIS file) or by specifying a regularly shaped grid pattern. These are referred to as **Shapefile Grid Definitions** and **Regular Grid Definitions**, respectively. A **Regular Grid Definition** is used when you want to specify a grid that is regularly shaped (e.g., 12 x 12 km squares). A **Shapefile Grid Definition** can be used to create either grids that are regularly shaped or grids that match an irregular shape, like a political boundary. All

Shapefile Grid Definitions must contain an attribute table with a unique (i.e., non-repeating) column and row index. Column and row values should not exceed 1,000,000.

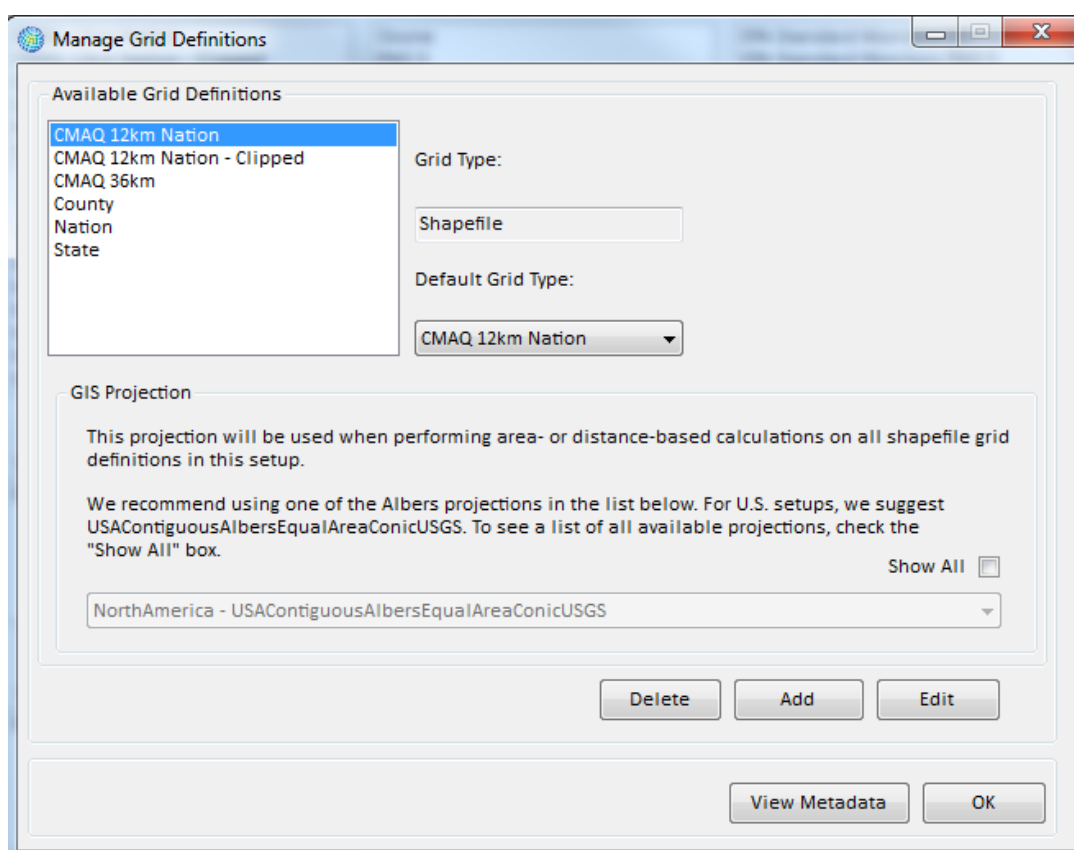
At least one **Grid Definition** should be created to outline the area of interest for the BenMAP-CE analysis (a city boundary, for example). Additional grid definitions can also be created for subdivisions of that area for which (a) data are available (see the Air Monitoring, Population, Incidence and Prevalence, and Variables sections below), or (b) reports or maps are desired.

For example, an analysis for the United States might use one or more of the following grid definitions:

- **Nation** – this Shapefile Grid Definition contains an outline of the United States (just the lower forty-eight states), defining an overall area of interest.

- **State** – this Shapefile Grid Definition contains state borders, for use in generating reports and maps with results aggregated to the state level.
- **County** – this Shapefile Grid Definition contains county borders, for use with county-based population and incidence rate data.
- **CMAQ 12km Nation** – this Shapefile Grid Definition contains grid cells that are roughly 12 kilometers on each side, for use with air quality modeling data.

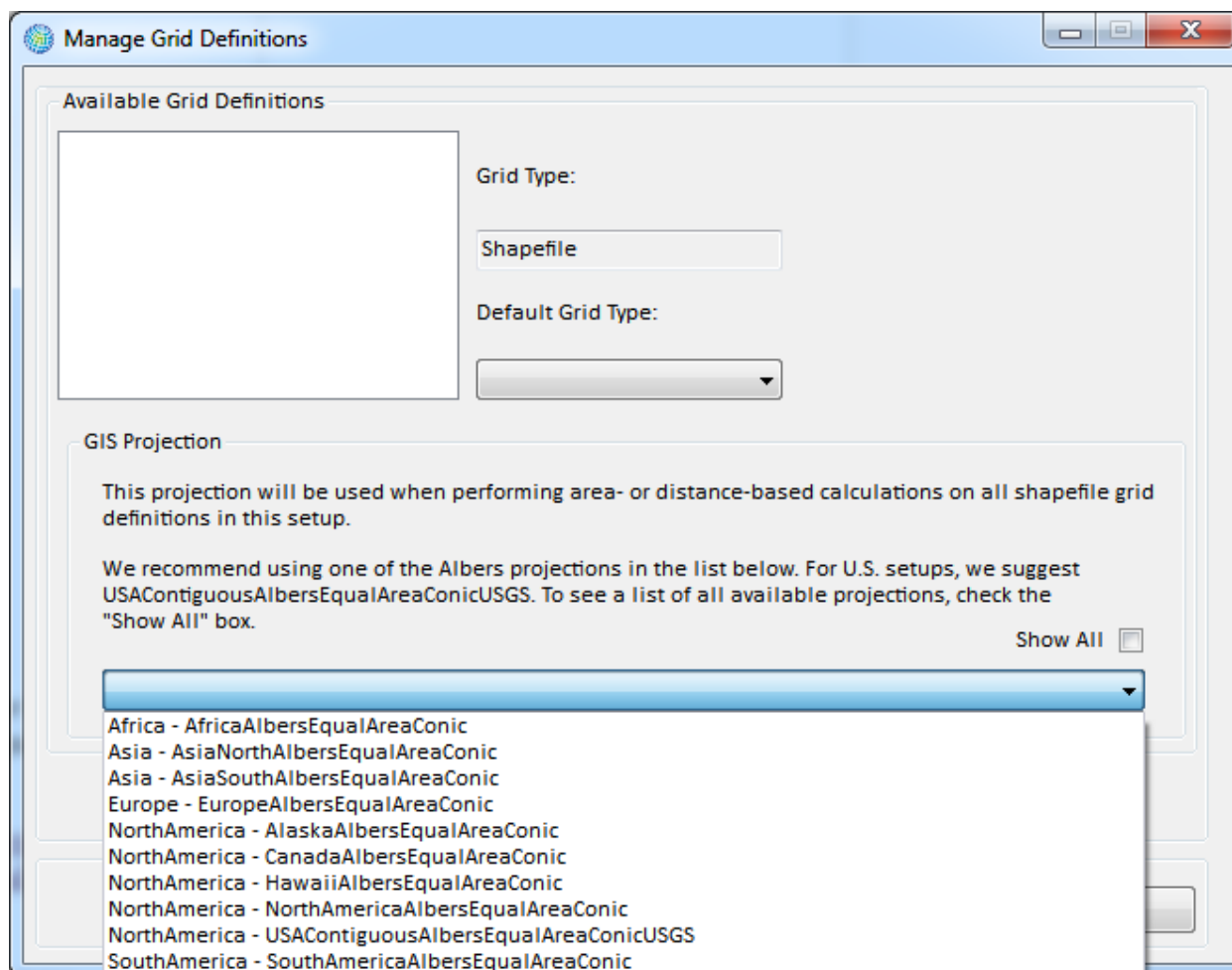
To start adding or modifying grid definitions, click **Modify Datasets** on the menu bar. Click on the **Manage** button below the **Grid Definitions** box. The **Manage Grid Definitions** window will appear.



Click on the **Add** button to display the **Grid Definition** window. Provide a name for the Grid Definition in the **Grid ID** field and specify the **Grid Type**: *Shapefile Grid* or *Regular Grid*.

When you add a new Setup, you will be asked to select a GIS projection. The projection you specify will be used when performing area- or distance-based calculations on all shapefile grid definitions in this setup. We recommend selecting a regional setting that best represents the geographic area you are evaluating from one of the Albers

projections provided in the list box. (For U.S. setups, we suggest USAContiguousAlbersEqualAreaConicUSGS.)



If you have reason to choose another type of projection (e.g., in consultation with a geospatial analyst), you may check the “Show All” box and select from an expanded list of choices.